MATH6006: Statistical Methods for OR Modelling

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Learning Outcomes

After this session, you should be able to

- Understand the central limit theorem (CLT) and its assumptions
- Apply CLT to approximate some repeated experiments
- Understand and apply the half-unit correction technique.

Motivating Example

A small example

Suppose we have a population of 7 secondary students and they got their monthly pocket money as follows (in £):

$$20,\ 40,\ 50,\ 60,\ 60,\ 75,\ 80.$$

The **population mean** is:

$$\mu = \frac{20 + 40 + 50 + 60 + 60 + 75 + 80}{7} = 55.$$

We would like to use the **sample mean** to infer the population mean. Can it be done?

If we sample each of the 7 students, we would get the true μ. However, it is often not possible to sample **whole** population.

Small Sample Size

Table: All samples of size $n = 2$

Sample $\{x_i\}$	Sample Mean $\overline{x}$	Sample $\{x_i\}$	Sample Mean $\overline{x}$
20, 40	30	50, 60	55
20, 50	35	50, 60	55
20, 60	40	50, 75	62.5
20, 75	47.5	50, 80	65
20, 80	50	60, 60	60
40, 50	45	60, 75	67.5
40, 60	50	60, 80	70
40, 60	50	60, 75	67.5
40, 75	57.5	60, 80	70
40, 80	60	75, 80	77.5
20, 60	40		

Histogram of Sample Means (Small Sample Size)

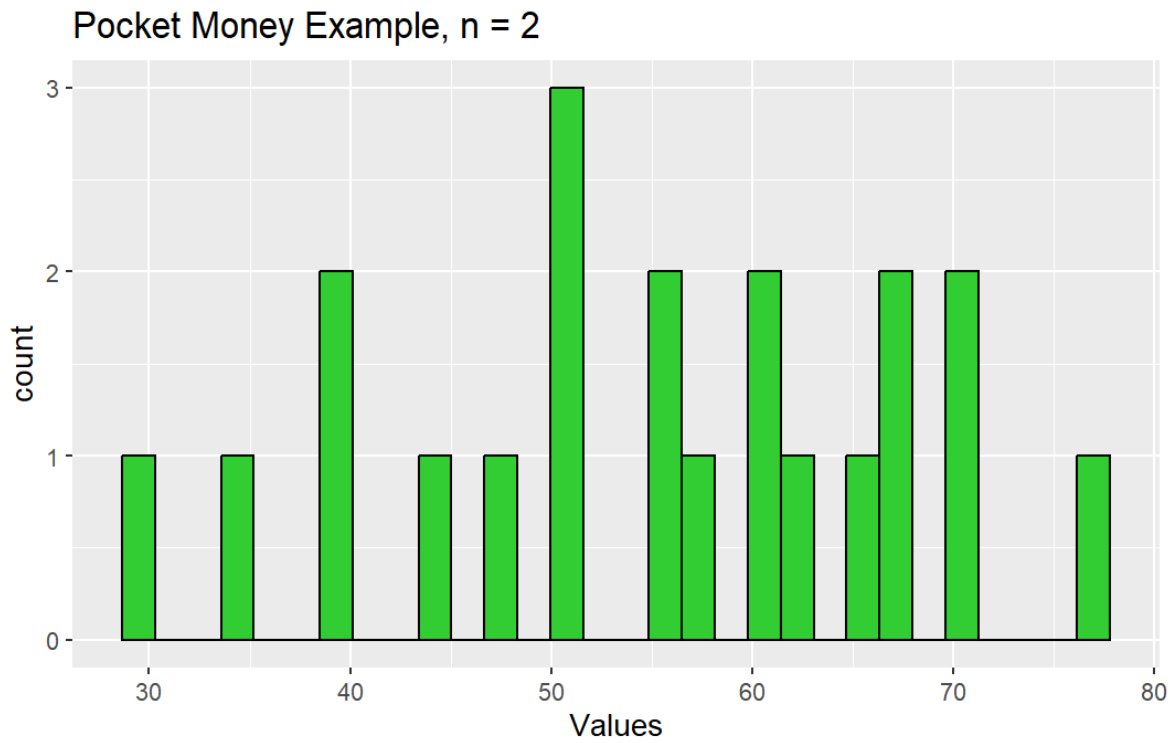


Figure: Distribution of Sampling Means ($n = 2$) for Pocket Money Data

Large Sample Size

Table: All samples of size $n = 6$

Sample $\{x_i\}$	Sample Mean $\overline{x}$
20, 40, 50, 60, 60, 75	50.1
20, 40, 50, 60, 60, 80	51.6
20, 40, 50, 60, 75, 80	54.1
20, 40, 50, 60, 75, 80	54.1
20, 40, 60, 60, 75, 80	55.8
20, 50, 60, 60, 75, 80	57.3
40, 50, 60, 60, 75, 80	60.8

Histogram of Sample Means (Large Sample Size)

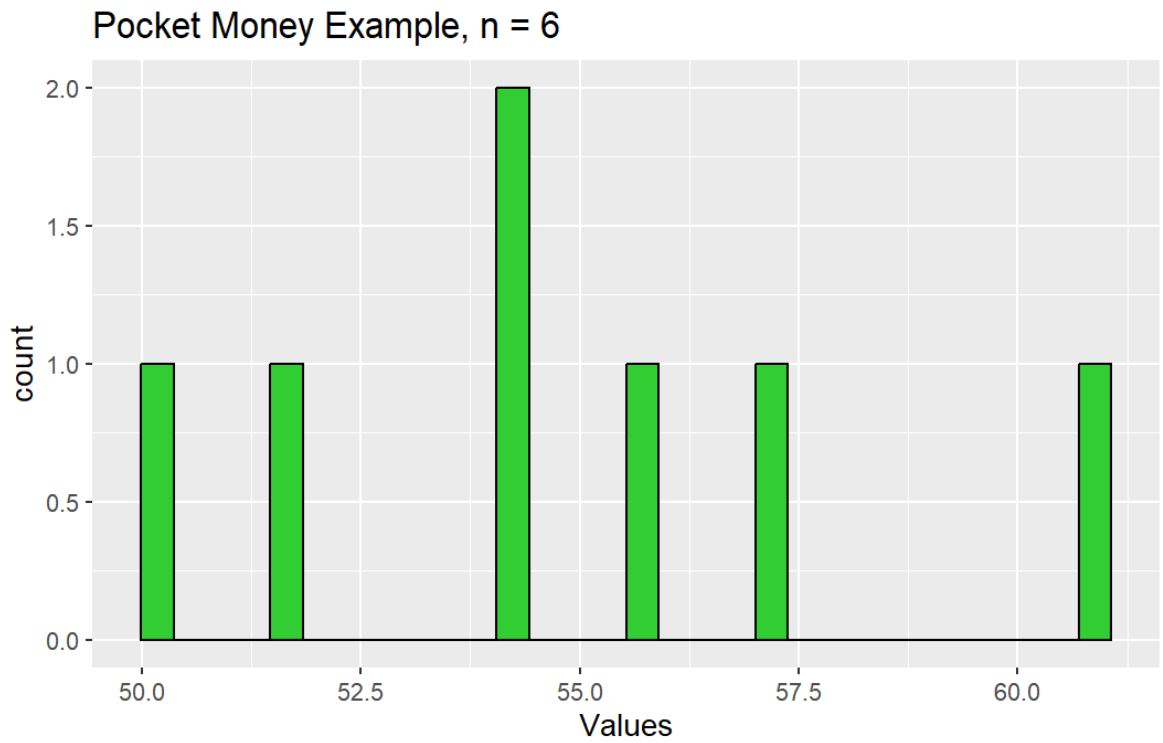


Figure: Distribution of Sampling Means ($n = 6$) for Pocket Money Data

Sample Mean

Definition

Let the random variables $X_1, X_2, \ldots, X_n$ denote a random sample from a population with the mean μ and the standard deviation σ. The **sample mean** value of these random variables is defined as:

$$\overline{X} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

Since X_i's are random, so is $\overline{X}$. Hence, we can talk about the mean and the variance of $\overline{X}$.

Sample Mean

We relate the mean of the samples to the population mean.

Sample Mean

The mean of $\overline{X}$ is the population mean μ because

$$E[\overline{X}] = \frac{1}{n}E[\sum_{i=1}^{n} X_i] = \frac{1}{n}\sum_{i=1}^{n} E[X_i] = \frac{1}{n}\sum_{i=1}^{n} \mu = \mu.$$

This does not depend on the sampling size n.

The Central Limit Theorem (CLT)

CLT

Let $X_1, X_2, \ldots, X_n$ be a set of n independent random variables having identical distributions with the population mean μ and the variance σ^2 (i.e., X_i's are i.i.d.). Let $\overline{X}$ be the mean of these random variables. Then, as n becomes large, the random variable

$$Z_n = \frac{\overline{X} - E[\overline{X}]}{\sigma_{\overline{X}}} = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}}$$

approaches the standard normal distribution $N(0, 1)$.

Law of Large Numbers

Let $X_1, X_2, \ldots, X_n$ be a set of n random variables having identical distributions with the population mean μ. Then

$$\overline{X} \to \mu \qquad \text{as} \qquad n \to \infty.$$

Comments on CLT

- The random variable Z_n depends on the number of n and is a **standardized random variable** such that

$$E[Z_n] = 0 \qquad \text{and} \qquad Var(Z_n) = 1.$$

But if the underlying population is normal, then Z_n is also a standard normal.
Using the fact: The sum of two independent normal distributions is still normal.

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Using the fact: The sum of two independent normal distributions is still normal.

- The Central Limit Theorem is used loosely in the following way:

$$\overline{X} \sim N(\mu, \sigma^2/n) \quad \text{for large n}$$

$$\sum_{i=1}^{n} X_i \sim N(n\mu, n\sigma^2) \quad \text{for large n.}$$

Large Example

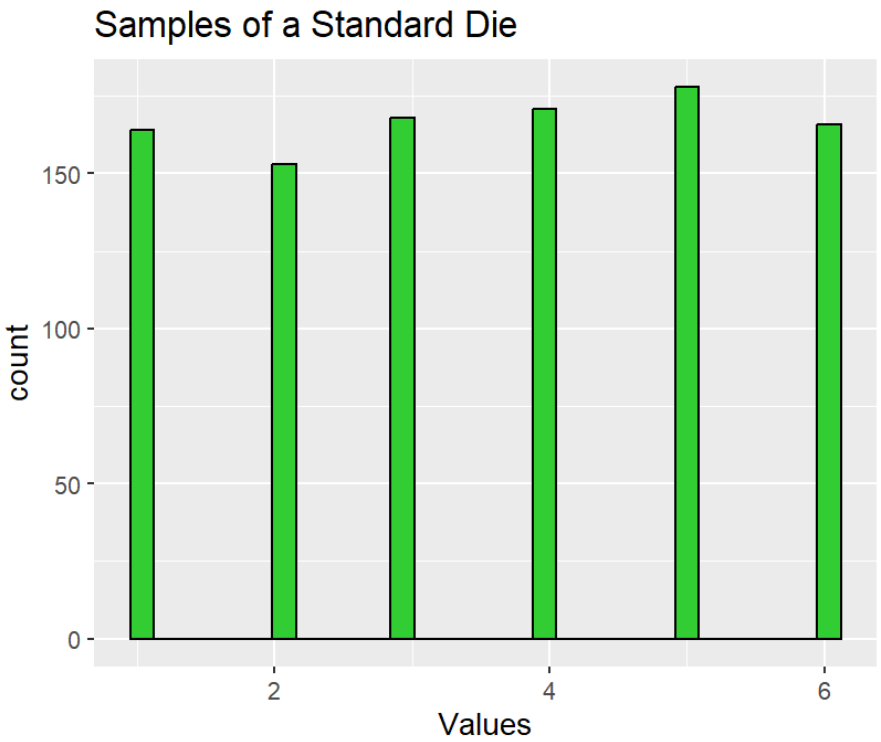


Figure: Nearly uniform distribution of face values of a die over 5000 random rollings

Histogram of Sample Means of Many Samples

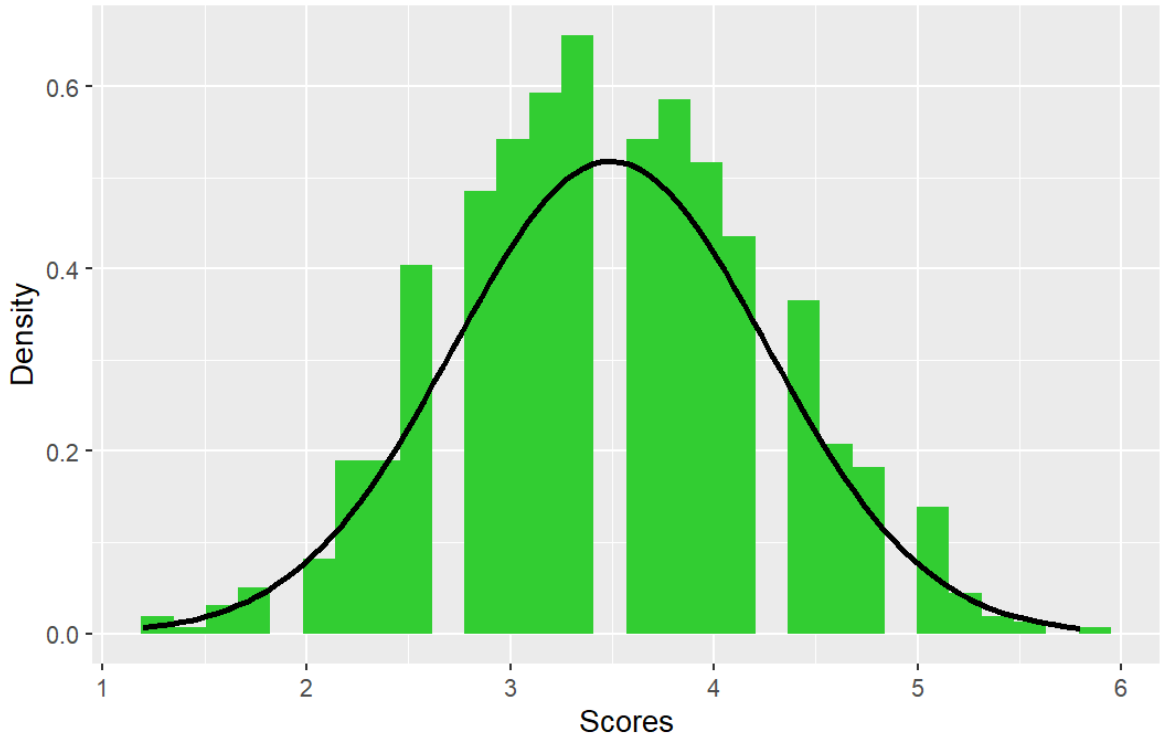


Figure: Distribution of Sampling Means of 1000 Samples (sample size = 5) from 5000 random rollings of a six-sided die

Sample Variance

We can relate the variance of the sample means to the population variance.

Theoretical Sample Variance

Assuming that X_i are independently, identically distributed (i.i.d.), then

$$\sigma^2_{\overline{X}} = \frac{\sigma^2}{n} \qquad \text{or} \qquad \sigma_{\overline{X}} = \frac{\sigma}{\sqrt{n}}.$$

$\sigma_{\overline{X}}$ is known as **Theoretical Sample Standard Error**.
This does depend on the sampling size n.

More on the Sample Variance

If we sample X with the values $x_1, \ldots, x_n$, then s^2 would be a good estimate for σ^2, where

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})$$

and $\bar{x}$ is the sample mean

$$\bar{x} = \frac{1}{n}(x_1 + \ldots + x_n).$$

When the underlying population follows a normal distribution, then we can say more about the distribution of S^2.

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When the underlying population follows a normal distribution, then we can say more about the distribution of S^2.
Suppose $X_1, \ldots, X_n$ are a random sample from a r.v. X, which has a normal distribution with the population variance σ^2. Then the new random variable

$$Y = \frac{(n-1)S^2}{\sigma^2} = \frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{\sigma^2}$$

follows the distribution χ^2_{n-1}.

Examples

Example (Example: Exponential Distribution)

Suppose $X_1, X_2, \ldots, X_n$ with $n = 25$ are i.i.d, each following $\text{Exp}(\lambda)$ with $\lambda = 1/100$. Use the Central Limit Theorem to estimate $P(90 \leq \overline{X} \leq 110)$, where $\overline{X}$ is the mean of the sample:

$$\overline{X} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

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It follows from the equations for the sample mean and sample variance that the expectation of the sample mean is

$$E[\overline{X}] = E[X_i] = \frac{1}{\lambda} = 100$$

and the variance of the sample mean is

$$Var(\overline{X}) = Var(X_i)/n = \frac{1}{n\lambda^2} = 400.$$

Examples

It follows from the Central Limit Theorem that

$$
\begin{aligned}
&P(90 \leq \overline{X} \leq 110) && (1)\\
&\approx P\left(\frac{90 - E[\overline{X}]}{\sqrt{Var(\overline{X})}} \leq \frac{\overline{X} - E[\overline{X}]}{\sqrt{Var(\overline{X})}} \leq \frac{110 - E[\overline{X}]}{\sqrt{Var(\overline{X})}}\right) && (2)\\
&= P\left(\frac{90 - 100}{20} \leq Z \leq \frac{110 - 100}{20}\right) && (3)\\
&= P\left(-\frac{1}{2} \leq Z \leq \frac{1}{2}\right) = 2\Phi(1/2) - 1 = 0.383. && (4)
\end{aligned}
$$

Examples

Example (Range of the Mean)

Suppose $X_1, X_2, \ldots, X_n$ with $n = 25$ are i.i.d from some distribution with the population mean $\mu = 100$ and the standard deviation $\sigma = 100$. Use the Central Limit Theorem to estimate $P(90 \leq \overline{X} \leq 110)$, where $\overline{X}$ is the mean of the samples:

$$\overline{X} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

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Example (Range of the Mean)

Suppose $X_1, X_2, \ldots, X_n$ with $n = 25$ are i.i.d from some distribution with the population mean $\mu = 100$ and the standard deviation $\sigma = 100$. Use the Central Limit Theorem to estimate $P(90 \leq \overline{X} \leq 110)$, where $\overline{X}$ is the mean of the samples:

$$\overline{X} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

The solution matches that for the first example as the CLT does not rely on whether or not the data came from the exponential distribution. We just need the mean and the variance.

Examples

Example (Southampton Football Club)

Southampton Football Club play 38 independent games every season. They have probability of 0.7 of winning each of the games. What is the probability they win at least 25 games?

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Note that we assume that each game is independent with the winning probability of $p = 0.7$. That is, $Y \sim bin(38, 0.7)$.
We need to calculate $P(Y \geq 25)$.

Examples

We can use the CLT to find $P(Y \geq 25)$.
First solution:

$$
\begin{aligned}
&P(Y \geq 25) \\
&\qquad \approx P\left(Z \geq \frac{25 - np}{\sqrt{npq}}\right) \\
&\qquad = P\left(Z \geq \frac{25 - 26.6}{\sqrt{7.98}}\right) \\
&\qquad = P\left(Z \geq \frac{25 - 26.6}{\sqrt{7.98}}\right) \\
&\qquad = P\left(Z \geq -0.5664\right) = P\left(Z \leq 0.5664\right) \approx 71.57\%.
\end{aligned}
$$

Examples

Second Solution:

$$
\begin{aligned}
P(Y \geq 25) &= P(Y \geq 25 - 0.5) \qquad \text{(“Continuity correction”)} \\
&\approx P\left(Z \geq \frac{24.5 - np}{\sqrt{npq}}\right) \\
&= P\left(Z \geq \frac{24.5 - 26.6}{\sqrt{7.98}}\right) \\
&= P\left(Z \geq \frac{24.5 - 26.6}{\sqrt{7.98}}\right) \\
&= P\left(Z \geq -0.7434\right) = P\left(Z \leq 0.7434\right) \approx 77.04\%.
\end{aligned}
$$

Continuity Correction

The reason why we used the *continuity correction* is that we are using a continuous random variable to approximate a discrete random variable. This practice makes the approximation more "reasonable" or "justifiable".

Examples

Example (BMW Dealership)

BMW plans to sponsor a new dealer in the city of Xuzhou, China. Existing research suggests that the deal will be successful in cities of this size if the mean annual family income is above $120,000$ RMB (Chinese currency). It is also assumed that the standard deviation of income is $60,000$ RMB in Xuzhou.

A random sample of 100 people was obtained and the mean income of the sample is $128,200$ RMB. Does this sample provide evidence to conclude that a dealer should be sponsored?

Examples

The Central Limit Theorem implies that the sample mean follows a normal distribution. This enables us to estimate the probability of the sample mean bigger than $128,000$ RMB if the population mean is $\mu = 120,000$.

$$
\begin{aligned}
&P(\overline{X} \geq 128000) \\
&\approx P\left(\frac{\overline{X} - E[\overline{X}]}{\sqrt{Var(\overline{X})}} \geq \frac{128000 - E[\overline{X}]}{\sqrt{Var(\overline{X})}}\right) \\
&= P\left(Z \geq \frac{128000 - 120000}{\sigma/\sqrt{100}}\right) \\
&= P\left(Z \geq 1.33\right) = 1 - \Phi(1.33) = 1 - 0.9082 = 0.0918.
\end{aligned}
$$